

Module 04: Cognition in Embodied

Learning Objectives

1. Define **Cognition** within the context of Embodied.
2. Analyze how Cognition interacts with other components of the Active Inference framework.
3. Apply specific constraints of Embodied to the formal definition of Cognition.

Introduction

This module explores **Cognition**. In the **Embodied** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Cognition is a critical component of the 8-part Active Inference spine, bridging the gap between Perception and Action.

Key Concepts

1. Cognition as a Markov Blanket Boundary

How does Cognition define the boundary between the agent and the environment?

2. Generative Models of Cognition

What parameters involved in Cognition must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Cognition drive the perception-action loop?

Applications

In Embodied, we see Cognition manifest in: * **Specific Example 1:** A mathematician reports that the proof "feels right" before she can articulate the formal steps -- a sense of aesthetic coherence that manifests as a warming expansion in her chest; this is intuitive cognition at work, where the generative model has compressed years of mathematical pattern recognition into embodied priors so deep that the evaluation of logical structure is experienced as a somatic judgment rather than a step-by-step deduction. * **Specific Example 2:** An experienced therapist notices a subtle tightening in her own diaphragm while listening to a client describe a seemingly positive event, and she intuitively cognizes that the client's cheerful narrative conceals grief; this embodied cognitive process is the therapist's generative model detecting a mismatch between the client's verbal content and the somatic signals crossing the interpersonal Markov blanket, with the prediction error registering as an intuitive insight in her own body before it becomes a conscious clinical hypothesis.

Conclusion

Understanding Cognition allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.